

Optical propagation

Optical fibre

- Uses the total internal reflection phenomenon
光纤利用**全内反射(total internal reflection)**现象来传输光信号。
- Typical path loss is 0.2 dB/Km 分贝 (decibel, dB)
 - Signal can go 150 Km without amplification

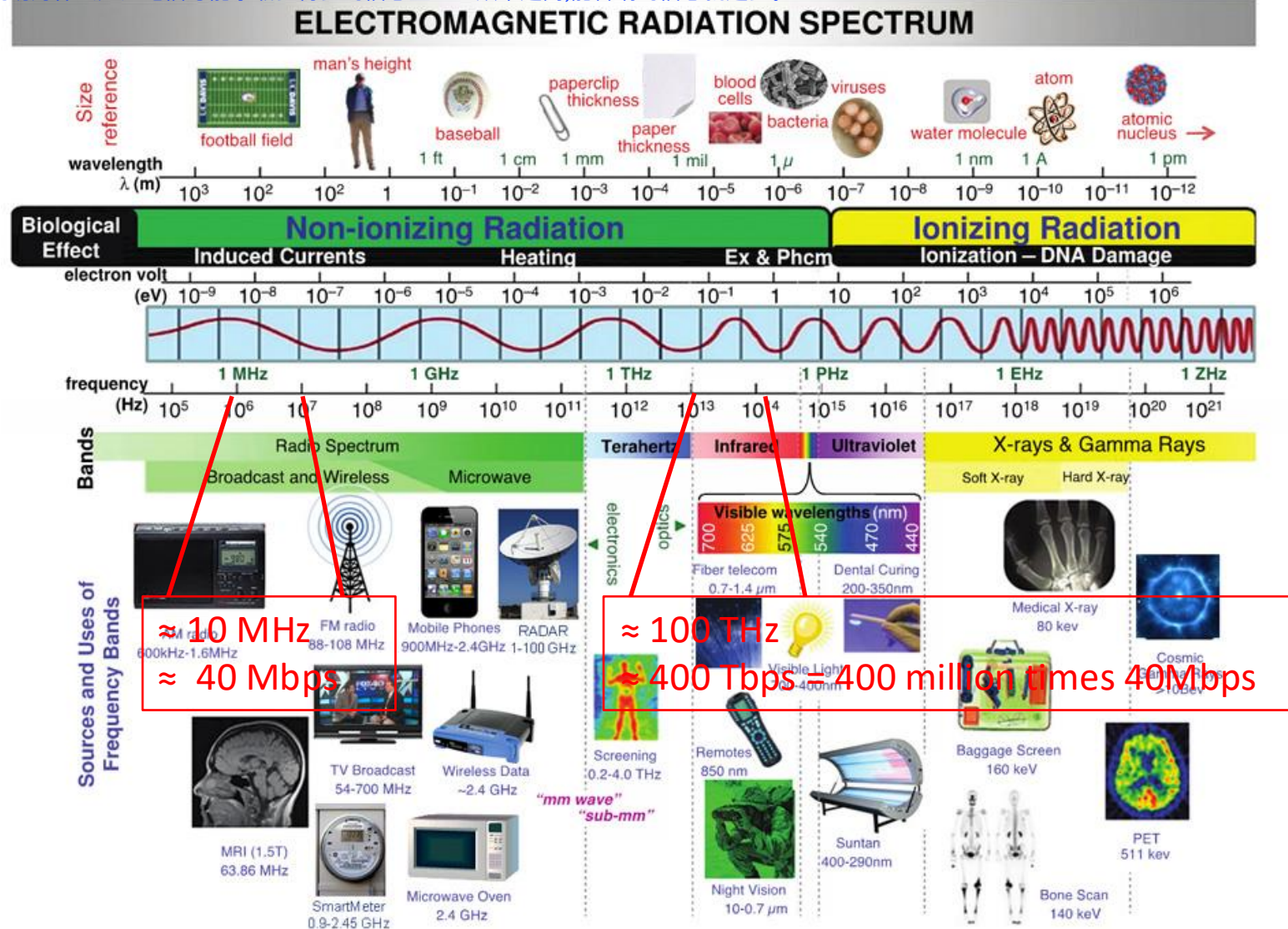


这张图展示了整个电磁波谱,从无线电波到伽马射线。光纤通信用的是红外光波段(大约100 THz频率)。对比一下带宽差距有多大:

无线电波(比如FM广播):约 10 MHz,能传约 40 Mbps
光纤用的频率:约 100 THz(比无线电高一万倍)
对应容量:约 400 Tbps —— 是前者的 4亿倍!

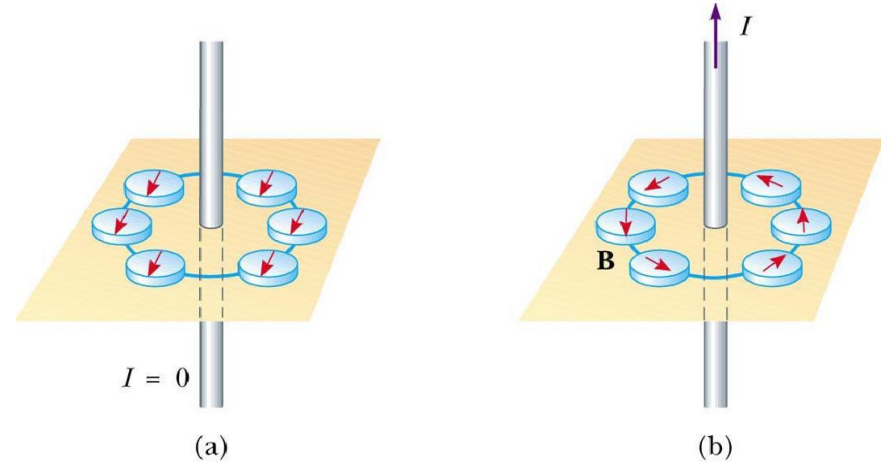
What is light?

这说明为什么光比电信号能承载多得多的信息量——频率越高,能容纳的信息就越多。



Radiation of copper wires

- A static current on a wire produces a static electro-magnetic field



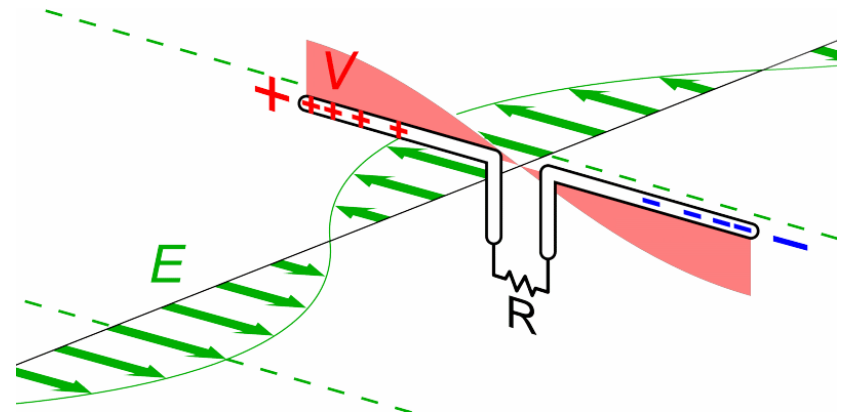
铜线的辐射问题(Radiation of copper wires)

铜线里静止不变的电流只会产生静态的电磁场(不会往外辐射能量)

但变化的电流(比如传输数据时的信号)会产生变化的电磁场,也就是说会向外辐射能量,造成能量损失和干扰

这是铜线传输的一个天生缺点。

- A variable current on a wire produces a varying electromagnetic field i.e., it radiates energy



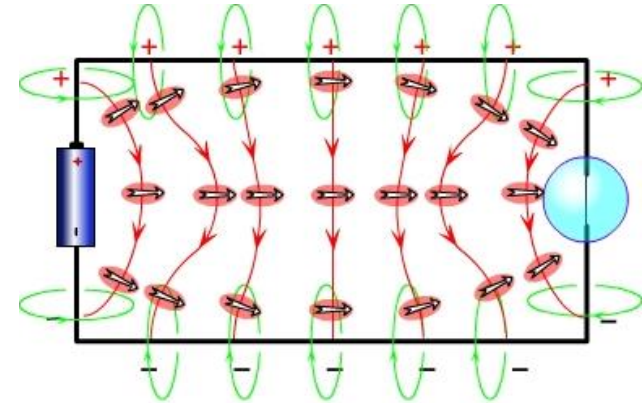
Wires and interference

电线和干扰(Wires and interference)

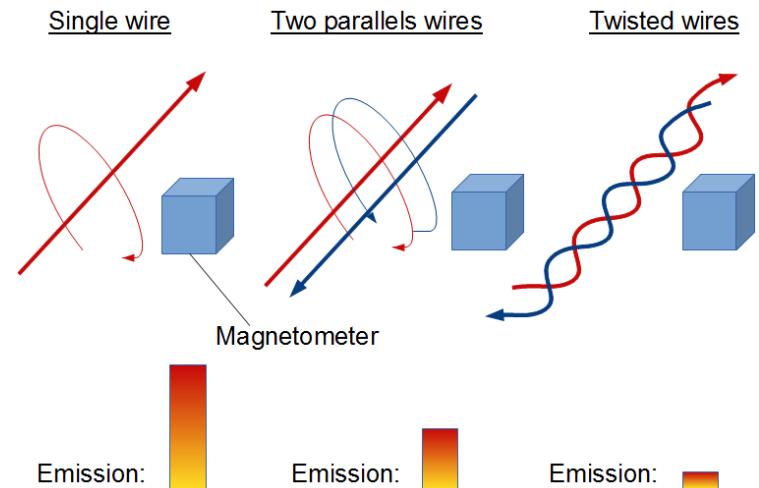
一对闭合电路的铜线,既会发射辐射,也会接收辐射(这叫"对偶定理"——天线发射和接收能力是对等的)。

****好消息:****去程电流和回程电流方向相反,它们产生的辐射会互相抵消一部分干扰。这也是为什么后面会讲"双绞线"的设计原理。

- A pair of copper wires closed into a circuit will both emit and receive radiation (the duality theorem says that an antenna is as good in transmitting as it is in receiving).



- Since going and returning current have opposite sign they tend to cancel each other interference



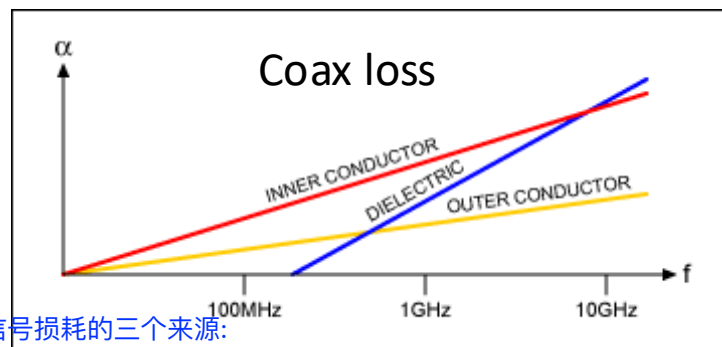
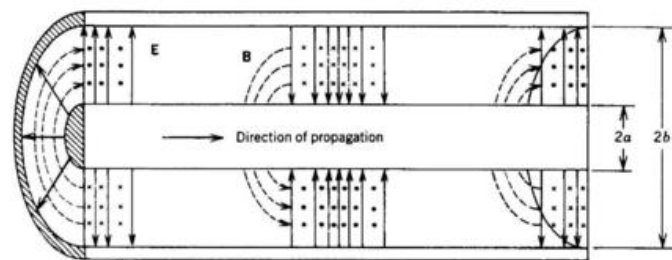
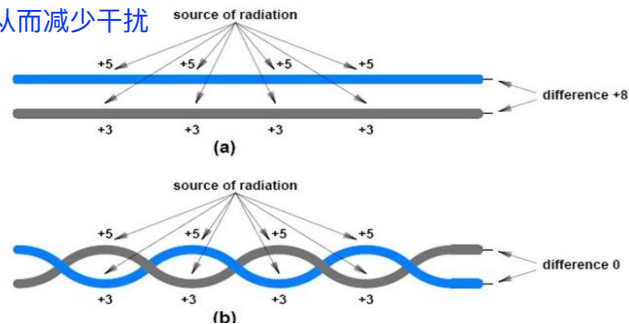
Twisted pair and coaxial

双绞线和同轴电缆(Twisted pair and coaxial)

双绞线:把两根线绞在一起,可以让外部干扰在两根线上产生的影响互相抵消,从而减少干扰

同轴电缆:利用法拉第原理,外层金属网把辐射限制在电缆内部,不让他跑出去

- Twisting the copper helps reducing interference from external sources
- The coaxial uses the Faraday principle by using an external sheet to confine the radiation within the cable
- Signal losses occur in (1) dissipation in the conductor, (2) dissipation in the dielectric and (3) radiation (small).
 - Most loss is in the conductor and at higher frequencies is lower for coax due to the larger conductor size (especially the external one)
 - Dielectric losses are also lower in coax as the inner material has typically lower density



信号损耗的三个来源:

导体本身的损耗(占大部分)

绝缘介质的损耗

辐射损耗(占比很小)

同轴电缆因为导体更粗(尤其外层),在高频时损耗反而比双绞线低。

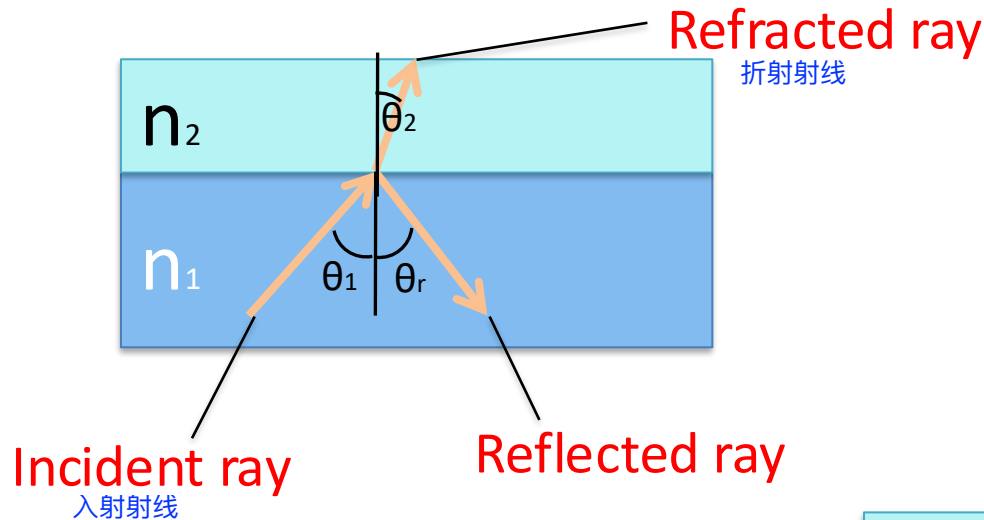
Optical fibre:

Total internal reflection

Consider two materials with different refractive index: n_1, n_2
折射率(refractive index)

$$n = \frac{c_0}{c}$$

c_0 → Speed of light in the vacuum
 c → Speed of light in the medium considered



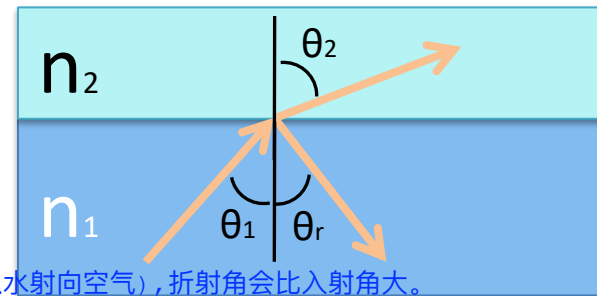
Snell's law:

斯涅尔定律

$$\begin{aligned}
 & - \theta_1 = \theta_r \\
 & - \frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1}
 \end{aligned}$$

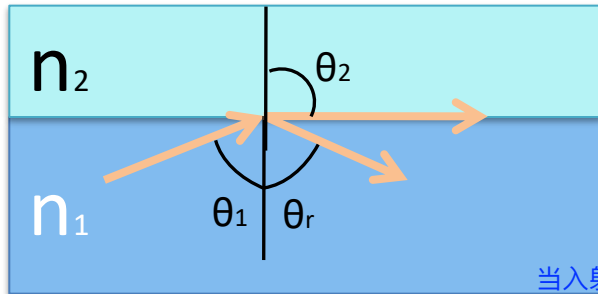
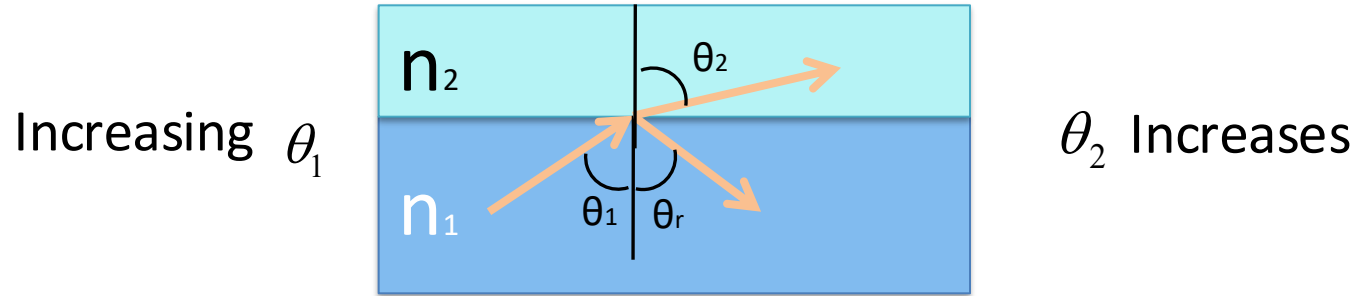
$$\text{If } n_1 > n_2 \quad \square \quad \theta_2 > \theta_1$$

Air
 Water



当光从一种介质 (n_1) 射向另一种介质 (n_2) 时, 如果 $n_1 > n_2$ (比如从水射向空气), 折射角会比入射角大。

Total internal reflection 全内反射



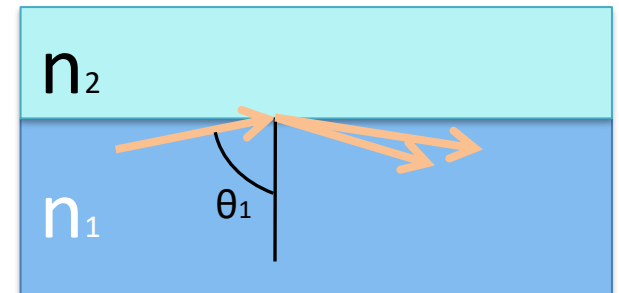
There is a value for $\theta_1 < \frac{\pi}{2}$ for which $\theta_2 = \frac{\pi}{2}$
 critical angle $\theta_{1-crit} = \sin^{-1}\left(\frac{n_2}{n_1}\right)$

当入射角 θ_1 逐渐增大, 折射角 θ_2 也会增大。存在一个临界角(critical angle)
 当入射角大于等于这个临界角时, 光就不会折射出去了, 而是完全反射回介质内部——这就是"全内反射"

这正是光纤工作的核心原理: 光被"锁"在纤芯里, 沿着纤芯一路反射前进, 几乎不漏出去。

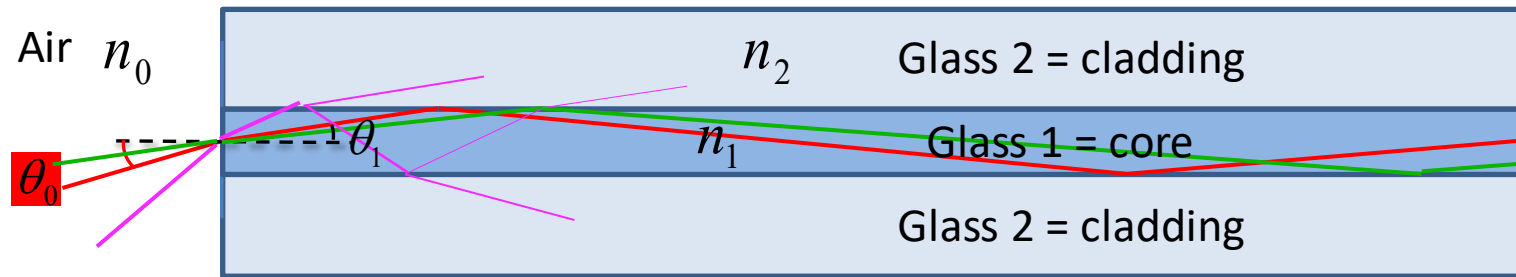
For $\theta_1 \geq \theta_{1-crit}$ we have total internal reflection
 The light remains confined inside n_2 !

This is how optical fibre operates



Propagation in optical fibre

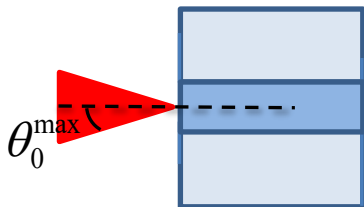
光纤中的光传播



光纤结构是:空气(n_0)→ 包层Cladding(n_2)→ 纤芯Core(n_1)→ 包层(n_2)

This fibre is called step-index, because the index n changes suddenly between cladding and core. 这是阶跃折射率光纤(step-index fibre)——因为纤芯和包层之间的折射率是突然变化的(不是渐变的)。

- Rays —— incident at an angle $\leq \theta_0^{\max}$ will propagate in the fibre through total internal reflection 只有以角度 小于等于 $\theta_0^{\max}$ 射入的光线,才能通过全内反射在光纤里传播
- Rays —— incident at an angle $> \theta_0^{\max}$ will loose power while propagating through the refracted ray 角度太大的光会"漏能量",逐渐消失



All rays with $\theta_0 \leq \theta_0^{\max}$ form a “Cone of acceptance”

所有满足条件的入射角度构成一个圆锥形区域,叫做**"接收锥"(Cone of acceptance)**:

$$\theta_0^{\max} = \sin^{-1}\left(\frac{\sqrt{n_1^2 - n_2^2}}{n_0}\right)$$

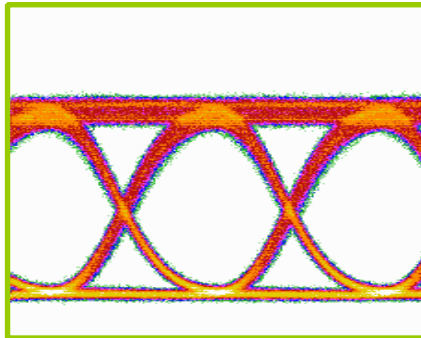
FIBRE FUNDAMENTALS

光纤基础四大问题(Fibre fundamentals)

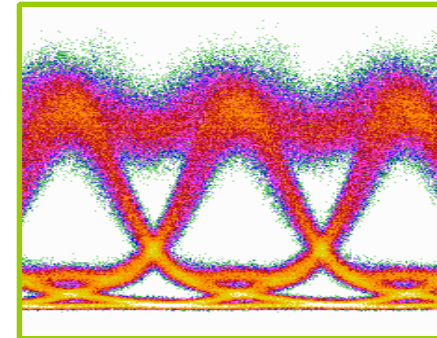
衰减(Attenuation)——功率随距离减少
色散(Dispersion)——脉冲变宽,不同频率/模式传播速度不同
非线性(Nonlinearity)——信号强度影响传播特性
偏振失真(Polarisation distortion)



The aim is to transmit signals with the minimum distortion



Transmitted Data Waveform



Waveform After 1000 Km

图中对比了"发送时的波形"和"传输1000公里后的波形"——可以看到眼图(那个X形状)变模糊了,说明信号已经严重失真,更容易出错。

目标:让信号传输时失真最小。

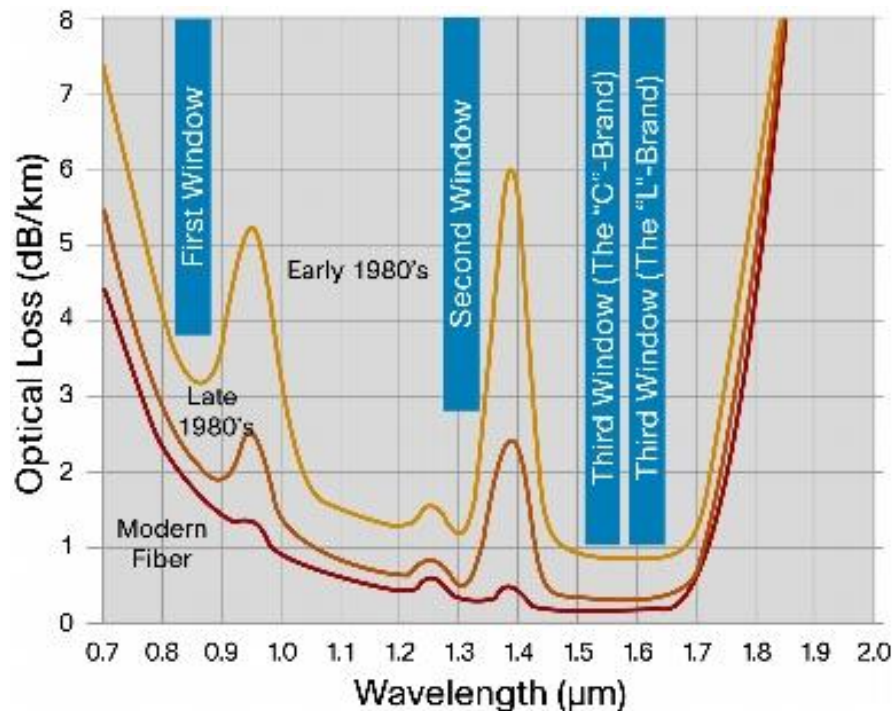
Path loss in fibre

光纤中的路径损耗(Path loss in fibre)

- The amount of power lost during propagation depends on the frequency (or wavelength) used for transmission.
- The smallest loss is at a wavelength of $1.55\text{ }\mu\text{m}$, and is 0.2 dB/Km

损耗大小取决于使用的波长。图表显示了不同年代光纤技术在各波长下的损耗曲线。

最小损耗出现在波长 $1.55\text{ }\mu\text{m}$ 处, 数值是 0.2 dB/km — 这就是现代光通信系统普遍选用这个波长的原因。



Dispersion in fibre

光纤中的色散(Dispersion in fibre)

- Dispersion in fibre is due to the fact that the refractive index n is not constant, but its value depends on the frequency of the signal.
- The dispersion coefficient D It is measured in $\frac{ps}{nm \cdot km}$
色散产生的原因是:折射率 n 不是固定值,而是随信号频率变化。
色散系数 D 的单位是: $ps/nm \cdot km$ (皮秒每纳米每公里)
- There are three types of dispersion:
 - Modal dispersion: because different modes propagate at different speed (only occurs in multi-mode fibre)
 - Chromatic dispersion: because different frequencies within a single mode propagate at different speed
 - Polarization mode dispersion: because different spatial polarization can propagate at different speed

三种色散类型:

模式色散(Modal dispersion)——不同"模式"(传播路径)速度不同,只发生在多模光纤

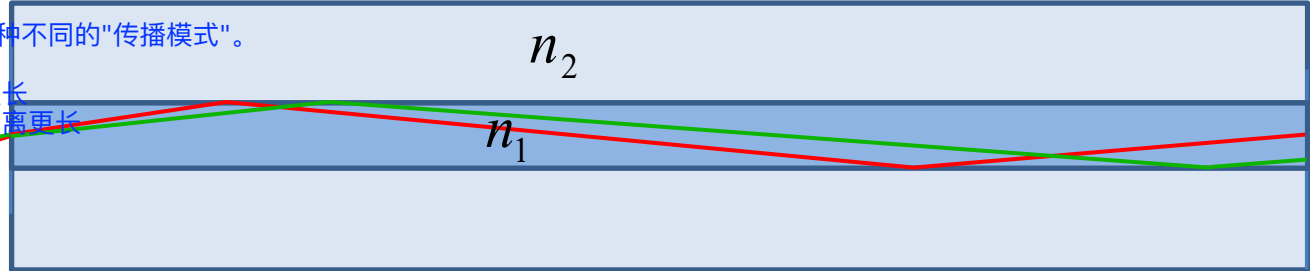
色度色散(Chromatic dispersion)——同一模式内,不同频率成分速度不同

偏振模色散(Polarization mode dispersion)——不同偏振方向速度不同

Modal dispersion in optical fibre

图中用绿色和红色两条光线代表两种不同的“传播模式”。

红色以更小的角度反射,走的路径更长
绿色和红色速度相同,但红色走的距离更长
所以红色会比绿色晚到达终点

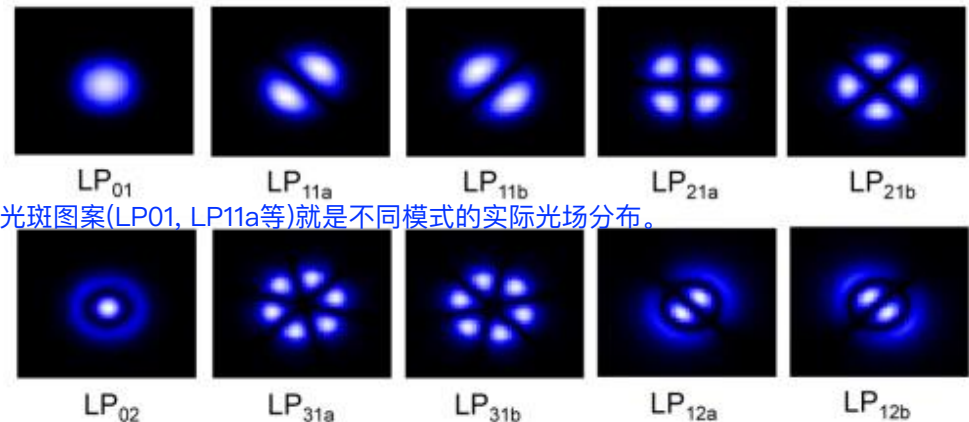


- The **green** and **red** rays are called modes of propagation in the fiber.
- Since **red** is reflected at a narrower angle than **green**, it will travel a longer distance for a same amount of fibre.
- Since **green** and **red** travel at the same speed, **red** will arrive at destination later than **green**.

More precisely, modes are solutions of the Maxwell equations applied to optical fibres.

更严谨地说,这些“模式”其实是麦克斯韦方程组在光纤中的解,图右侧那些光斑图案(LP01, LP11a等)就是不同模式的实际光场分布。

Visualising the different modes (in special coordinates x,y)



Impairments caused by modal dispersion

模式色散造成的损伤(Impairments)

- If I transmit a pulse it will be composed of a number of different modes, for simplicity here are indicated with two colors, red and green



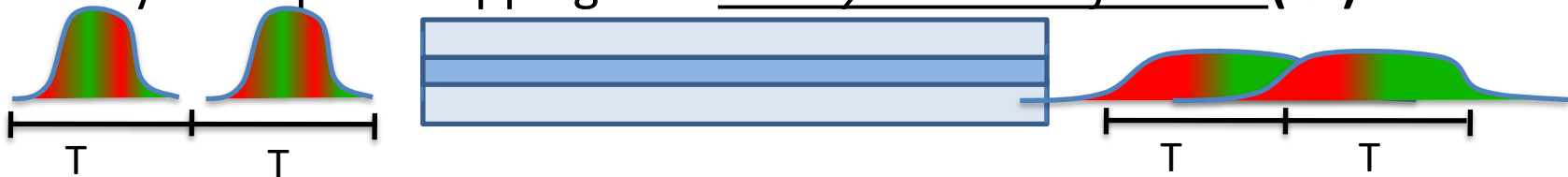
because **red** is slower than **green** the pulse spreads, and its peak power decreases

如果发送一个脉冲,它其实是由多种模式(简化为红、绿两种)组成的。

因为红色比绿色慢,脉冲会在传播过程中被拉宽,峰值功率下降。

如果连续发送一串脉冲,拉宽后的脉冲会互相重叠,这叫码间干扰 (intersymbol interference, ISI)
接收端很难分辨这是"1"还是"0",容易导致错误。

- In a communication channel where I transmit a series of pulses, they end up overlapping → intersymbol interference (ISI)



Single-mode vs. multi-mode

- A multi-mode fibre has a larger core, which allows multiple modes to coexist at the frequencies used for optical communication
- A single-mode fibre has a smaller core, which only allows one propagation mode.

单模 vs 多模光纤(Single-mode vs multi-mode)

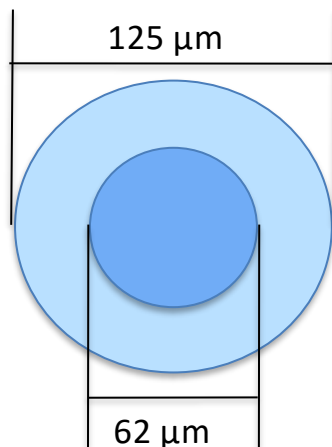
多模光纤:纤芯较大(比如62 μm),允许多种模式同时存在 → 有模式色散问题

单模光纤:纤芯很小(只有9 μm),只允许一种模式传播 → 完全消除了模式色散!

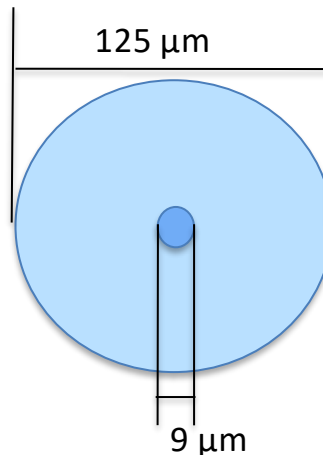
这就是为什么电信系统大多使用单模光纤。

- **Modal dispersion is simply eliminated by using single-mode fibre!!**

Multi-mode fiber



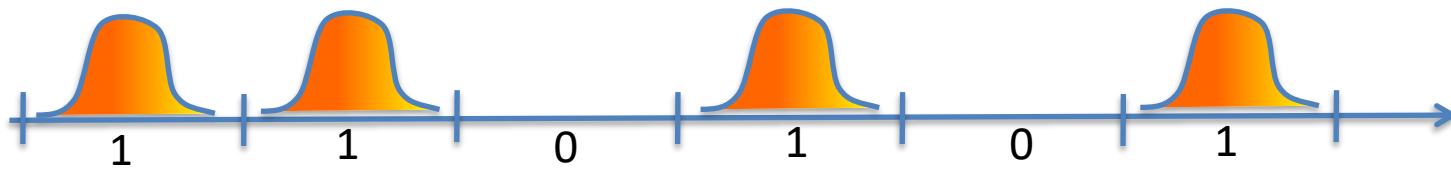
Single-mode fiber



Single-mode fiber is the most used in telecommunications

Chromatic dispersion problem

- Optical communications is achieved by transmitting pulses

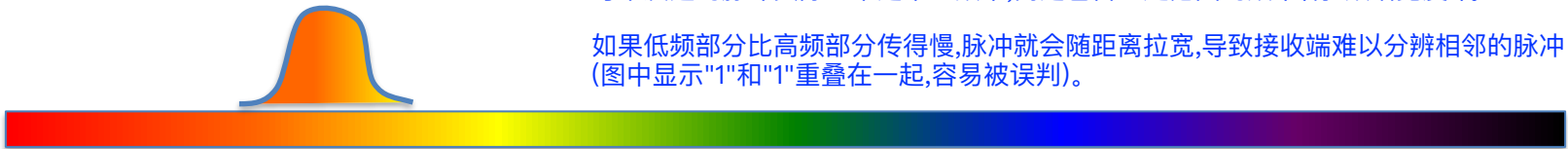


- Each pulse has a finite spectral width (it's made of a number of adjacent frequencies)

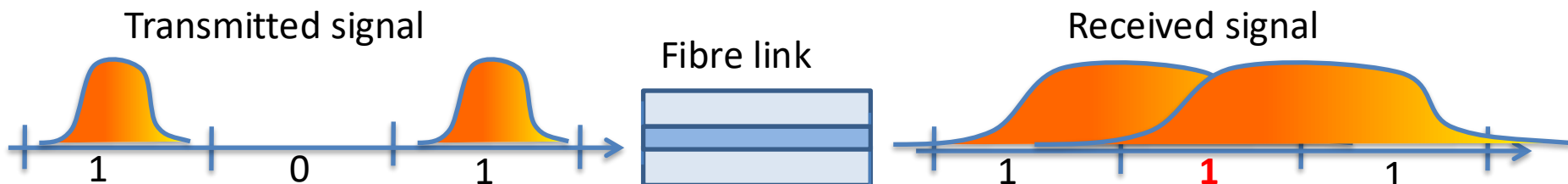
色度色散问题(Chromatic dispersion problem)

每个发送的脉冲实际上不是单一频率,而是包含一定范围的频率(有"频谱宽度")。

如果低频部分比高频部分传得慢,脉冲就会随距离拉宽,导致接收端难以分辨相邻的脉冲(图中显示"1"和"1"重叠在一起,容易被误判)。



- If lower frequencies travel slower than higher ones, the pulse spreads, causing transmission errors



Important note

- Dispersion is a linear phenomenon!

=> it cannot create new frequencies. 色散是线性现象,它不会产生新的频率成分

- In principle there is a time/frequency duality, so pulses that are shorter in time have larger bandwidth (in the frequency domain).

=>However, this is different for dispersion!

- A periodic signal can be decomposed as a sum of sin functions, of multiple frequency, each with given amplitude and phase.

□ 一个周期信号可以分解成多个正弦波的叠加(傅立叶级数):

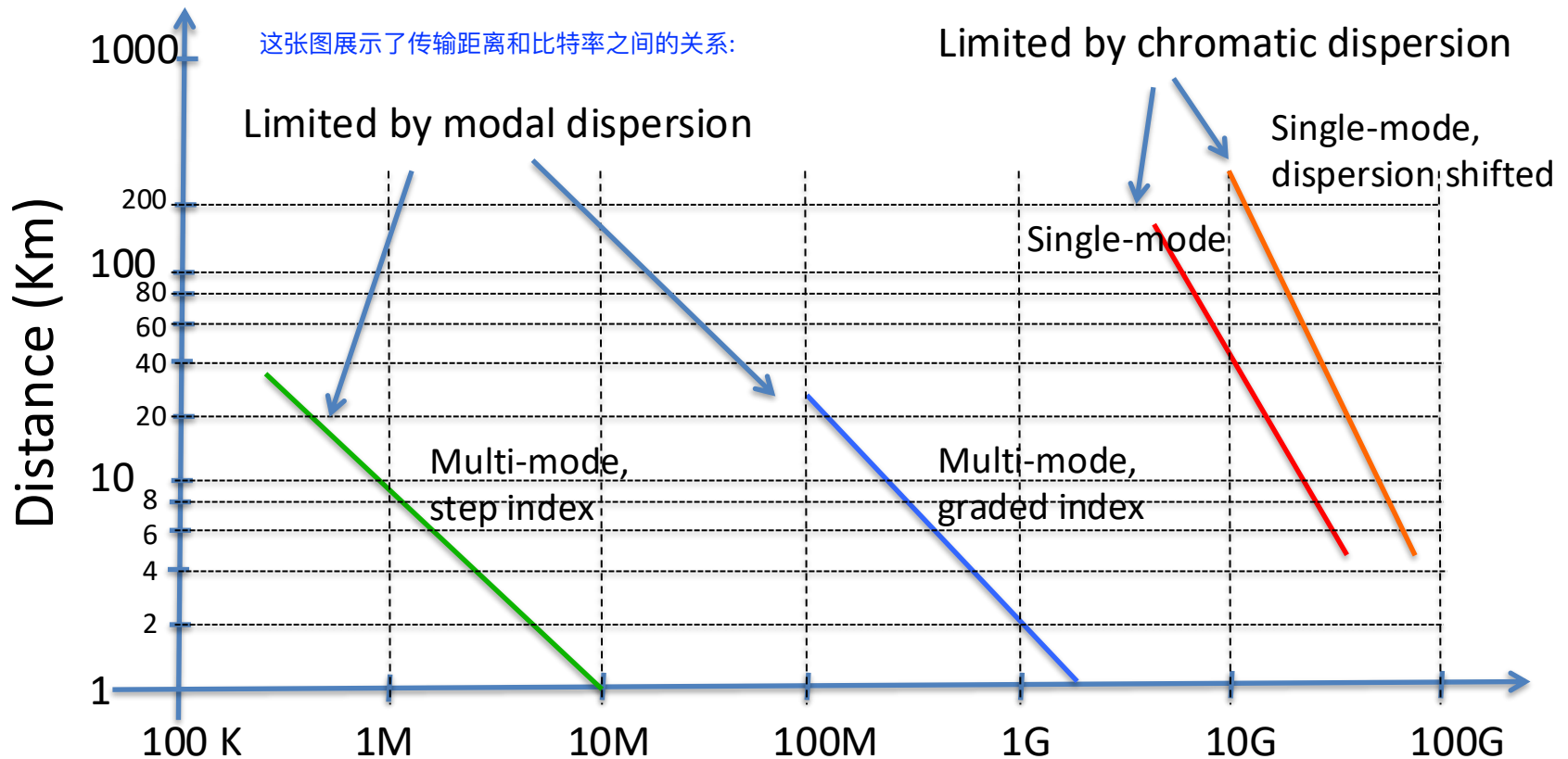
$$s(t) = C_0 + \sum_{n=1}^{\infty} C_n \sin(2\pi nft + \varphi_n)$$

- Dispersion does not create new frequencies, but it modifies the phase of the signal 色散不创造新频率,而是改变每个频率分量的相位
- As a result, when summed up (at the receiver or anywhere else in the fibre) the components will be out of phase, which makes the pulse broader in the time domain 当这些分量在接收端重新叠加时,因为相位不同步,导致脉冲在时间上变宽

Modal and chromatic dispersion

模式色散与色度色散对比图

- Modal dispersion **only** occurs in multi-mode fibres
- The amount of dispersion depends both on the data rate, and on the length of fibre where the signal propagates



多模光纤(阶跃折射率):受模式色散限制,距离/速率都最差
多模光纤(渐变折射率):稍好一些
单模光纤:受色度色散限制,可以支持更高速率、更长距离
单模+色散位移光纤:表现最好

结论:模式色散影响最大,单模光纤基本消除了这个问题,能跑得更远更快。

Bit rate (bps)

Modal and chromatic dispersion

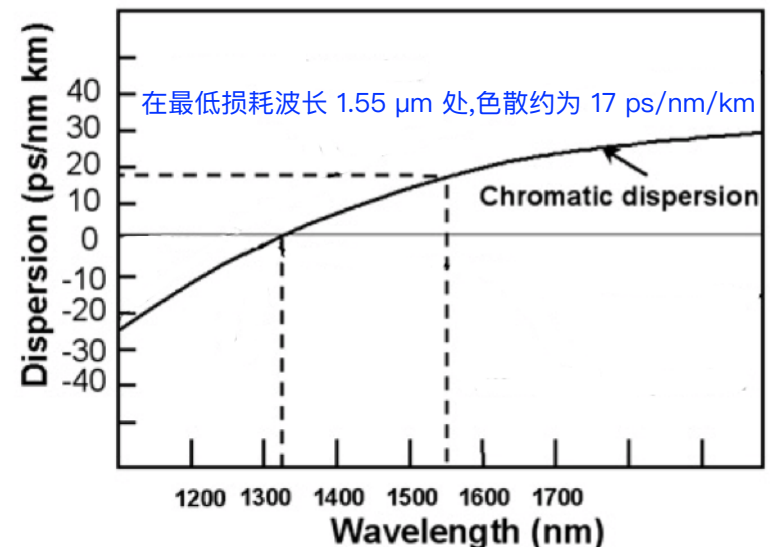
- Modal dispersion has the biggest effects:
 - **This is why telecoms systems use mostly single mode fibres!**
模式色散影响最大 → 这是电信系统都用单模光纤的原因
- Chromatic dispersion is the second most important cause:
 - Due to the fact that different frequency components of a pulse travel at different speed 色度色散是第二重要的因素
色度色散用"皮秒每纳米带宽每公里"衡量,意思是:脉冲带宽1纳米,传1公里,会展宽多少皮秒
 - Dispersion is measured in Picoseconds per nanometer of bandwidth per kilometer

➔ How many nanoseconds does a pulse broadens, if its bandwidth is 1 nanometer and it has travelled for 1 Km

- The lowest loss wavelengths of $1.55\ \mu\text{m}$ has 17ps/nm/km dispersion

在最低损耗波长 $1.55\ \mu\text{m}$ 处,色散约为 17 ps/nm/km
而在 $1.3\ \mu\text{m}$ 处,色散几乎为零

- While at $1.3\ \mu\text{m}$ dispersion is almost null



有趣的矛盾:损耗最小的波长($1.55\ \mu\text{m}$)色散反而较大,损耗较大的波长($1.3\ \mu\text{m}$)色散却很小——这就需要权衡取舍,或者用特殊设计的光纤来同时优化两者。

Dispersion shifted fibre

色散位移光纤

- Dispersion in fibre is due to:
 - the material (cannot be changed)
 - the shape of the waveguide, i.e. the shape of the fibre cross section (this can be changed, like we did for the graded-index fibre)

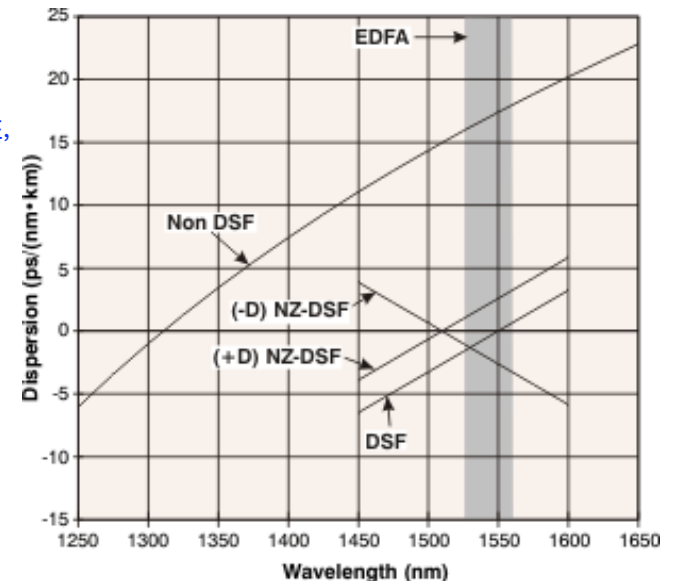
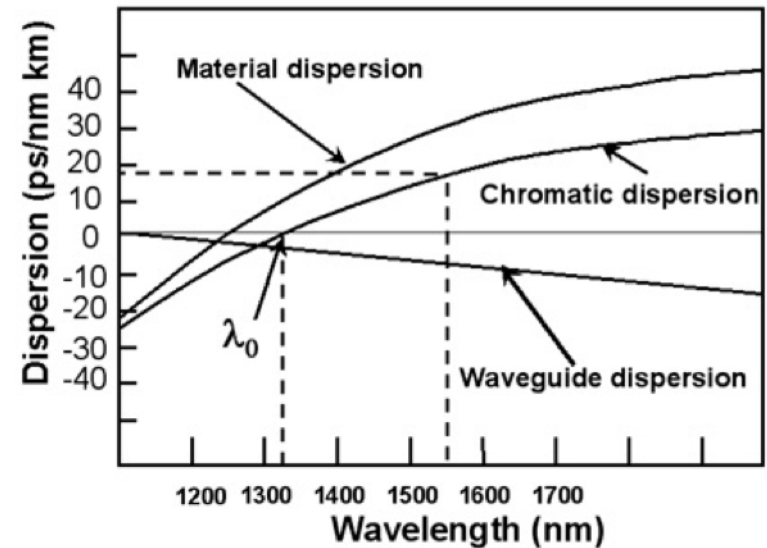
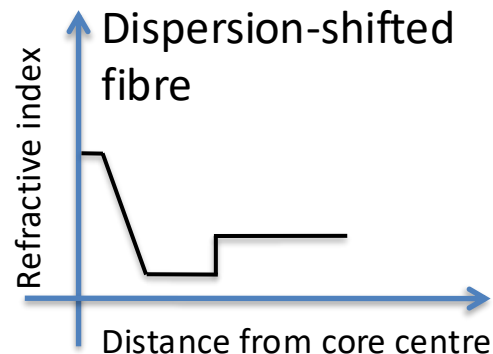
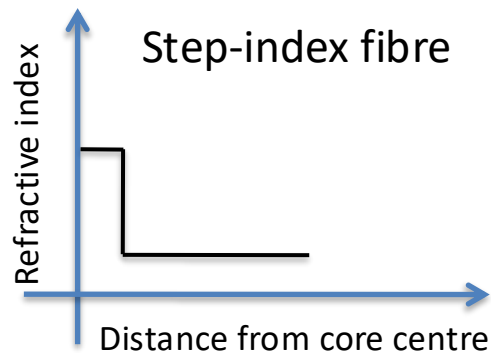
色散来源于两部分:

材料本身——无法改变

波导形状(光纤横截面的折射率分布形状)——可以人为设计和改变

- I can modify the dispersion by changing the distribution of the refractive index

通过设计折射率的分布方式(比如"阶跃型" vs "色散位移型"),工程师可以主动调整色散特性,让最低损耗波长和最低色散波长尽量对齐,提升整体性能。



Polarization-mode dispersion (PMD)

- The electric field in a single mode fibre propagates with two orthogonal polarization
- Because of imperfect production techniques, the fiber is not completely symmetrical

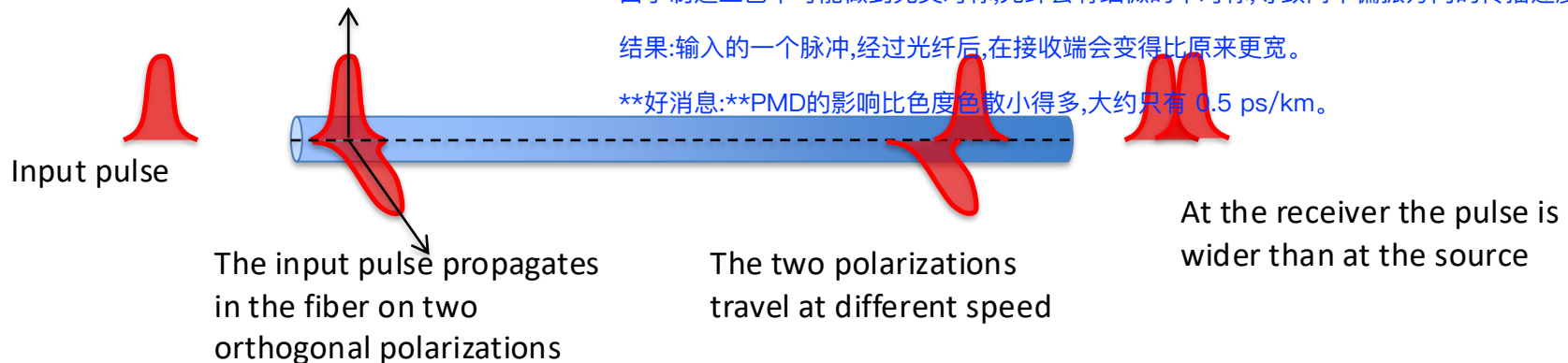
→ The two orthogonal components of the electric field have different propagation characteristics

单模光纤中,电场其实是以两个正交偏振方向传播的。

由于制造工艺不可能做到完美对称,光纤会有细微的不对称,导致两个偏振方向的传播速度略有不同。

结果:输入的一个脉冲,经过光纤后,在接收端会变得比原来更宽。

好消息:PMD的影响比色度色散小得多,大约只有 0.5 ps/km。



- PMD is less severe than chromatic dispersion: about 0.5 ps/km

Non-linear effects

非线性效应(Non-linear effects)

非线性意味着光纤的传输特性会随着信号本身的强度而变化——功率越大,非线性效应越明显。

- Nonlinearity implies that the transfer function of the fiber depends on the signal being transmitted. Their effects increase with the transmitted power.

几种主要非线性效应:

拉曼散射(Raman scattering):光与光纤材料中的“声子”相互作用,产生新频率的光

自发拉曼 → 产生随机噪声

受激拉曼 → 产生和原信号同相位的放大效果(可用来做“拉曼放大器”)

自相位调制(SPM):信号自身强度变化导致折射率变化,从而影响自己的传播

交叉相位调制(CPM):在波分复用(WDM,多个波长同时传输)系统中,一个波长的调制会影响其他波长

四波混频(FWM):三阶非线性效应会产生全新的频率(不同于色散,色散不产生新频率),频率是几个输入信号频率的加减组合。

这在WDM系统中会产生额外噪声,波长间距越小,问题越严重

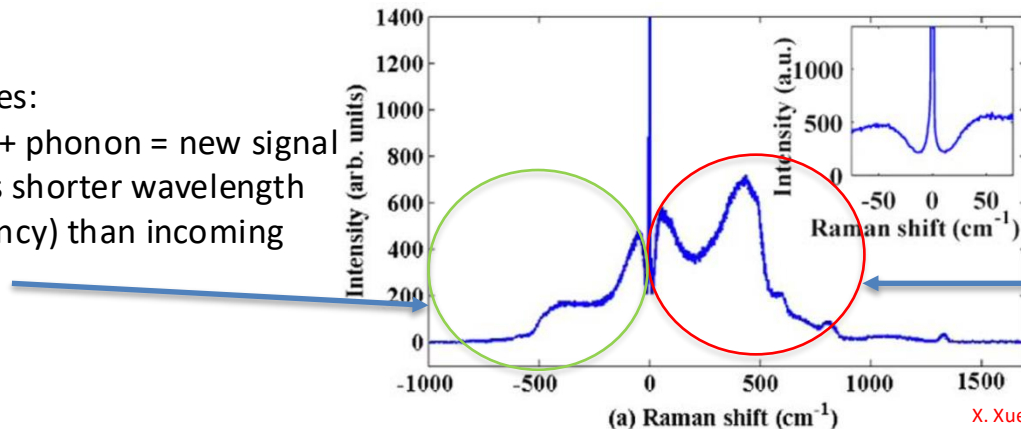
- Raman scattering:

共同点:SPM和CPM都会让脉冲变宽,相当于增加了额外的色散,在高速率系统中问题更突出。

- Light interacts with phonons, generating photons with different frequency from the incoming signal.
- Spontaneous → random frequency and phase = noise
- Stimulated → same frequency and phase = amplification (there are Raman amplifiers that make use of this effect)

Anti-Stokes lines:

existing signal + phonon = new signal
New signal has shorter wavelength
(higher frequency) than incoming
signal

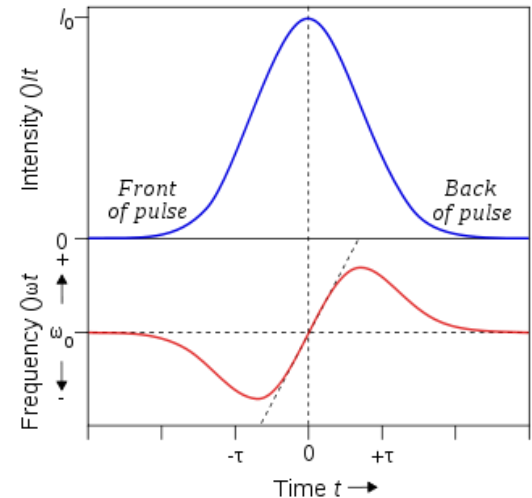


Stokes lines:

existing signal – phonon = new signal
New signal has longer wavelength
(lower frequency) than incoming
signal

Non-linear effects

- Self-phase modulation (SPM):
 - since the refractive index depends on the intensity of the traveling light, an amplitude-modulated signal sees different values of n , and is subject to a different propagation constant
- Cross-phase modulation (CPM): if there are multiple wavelengths in a fiber (WDM), the modulation of other wavelengths produce changes in the refractive index that affects (re-modulate) the signal

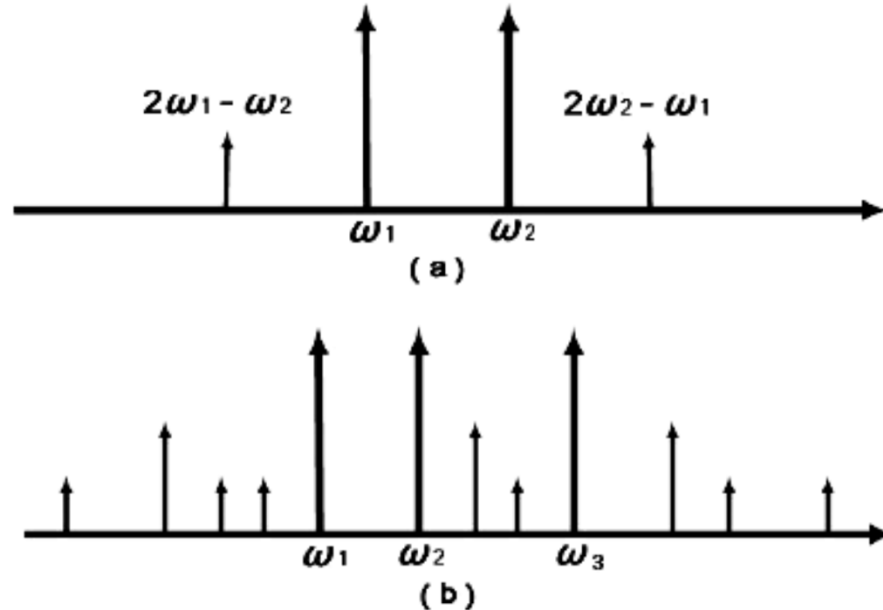


Same concept but the effect is across different signals

Both SPM and CPM produce a broadening of the pulse, so create additional dispersion and are more problematic at higher rates

Non-linear effects

- Four-wave mixing (FWM):
 - the refractive index nonlinearity also generates new frequencies
 - Based on 3rd order nonlinearity, produces signals whose frequency is the sum or difference of input signals (i.e., $\omega = \omega_i \pm \omega_j \pm \omega_k$)
 - This can generate noise in WDM systems. It does not depend on data rate, but it is worse when the wavelength spacing is small



Shannon capacity and nonlinearity

香农容量与非线性

$$C = B \cdot \log_2 \left(1 + \frac{S}{N} \right)$$

意思是:信道容量 C 正比于带宽 B , 以及信噪比(S/N)的对数。

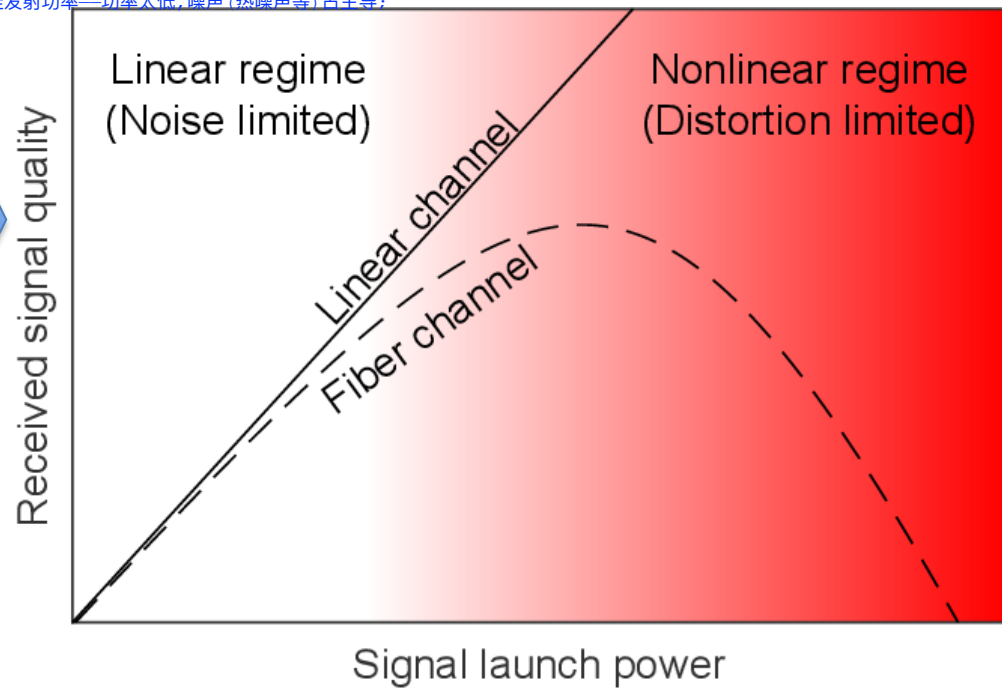
The capacity of a communication channel is proportional to the Bandwidth of the signal and to the $\log_2$ of the signal to noise ratio

→ In linear regime the higher the signal power, the higher the capacity

正常逻辑: 信号功率越大, 信噪比越高, 容量应该越高。

但是问题来了: 在光纤中, 非线性效应产生的噪声也随功率增大而增大! 所以实际上存在一个最佳发射功率——功率太低, 噪声(热噪声等)占主导; 功率太高, 非线性失真占主导。两者之间有个“甜蜜点”能让容量最大化。

However, we have seen that nonlinearity introduces noise that is proportional to the signal power



What are key features of nonlinear effects?

Mark all that apply

- A. They depend on the power of the signal
- B. They produce interaction across signals
- C. They only produce unwanted effect (i.e., noise)
- D. They are stronger at longer wavelengths

When considering the Shannon capacity limit
in optical fibres: Mark all that apply

- A. It does not not apply to optical communications
- B. It implies that capacity always increases with higher transmission power
- C. It implies there is an optimal transmission power to maximise capacity
- D. It implies that higher bandwidth always leads to higher capacity

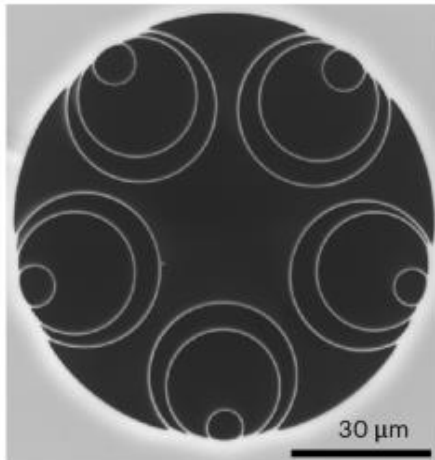
A look at the future: hollow-core fibres

未来展望:空芯光纤(Hollow-core fibres)

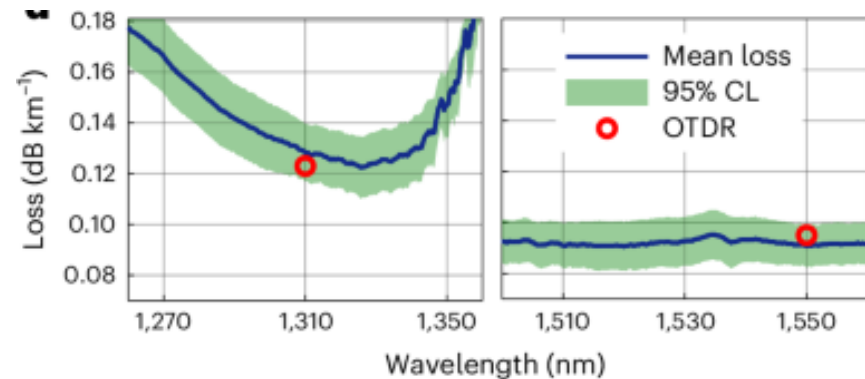
空芯光纤不用玻璃做纤芯,而是用空气或气体。

问题:空气的折射率比周围材料低,不能用传统的全内反射原理导光,所以需要用反谐振现象(anti-resonant phenomena)来引导光传播。

- A hollow core fibre does not use glass as core, but air or a gas
 - Issue is that refractive index of the core becomes less than that outside
 - ➔ It uses anti-resonant phenomena to guide the light



Cross-section showing fibre profile



Loss performance in the spectrum

Loss can go beyond standard fibre theoretical limits of 0.14dB/km

优势(相比标准单模光纤):

损耗更低 → 可以减少一半的放大器(EDFA)数量,降低成本
纤芯几乎没有非线性效应 → 可以用更高的发射功率、注入更多信道
延迟更低,接近真空中光速(因为光在空气中传播比在玻璃中快)
更大带宽、更低色散

Performance

劣势:

制造工艺复杂(精密几何结构)
熔接和连接器技术还不成熟
耐用性可能有问题(结构复杂、玻璃薄)

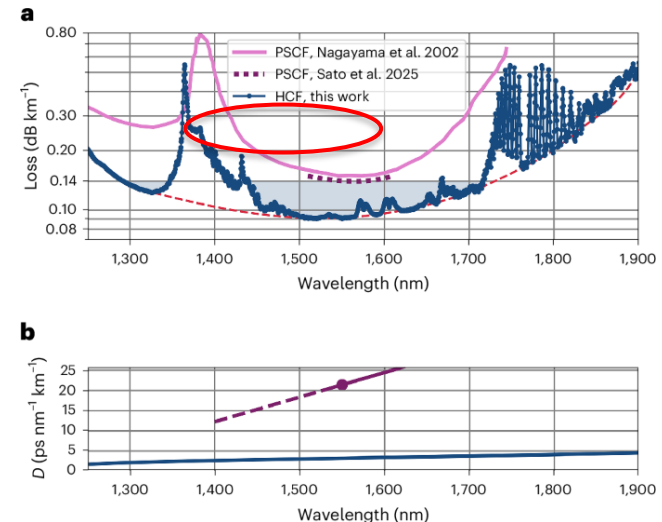
Advantages (compared to standard SMF)

- Lower loss means can reduce by half number of EDFAs (lower cost and better OSNR)
- The core has negligible nonlinearity (some just in the edge)
 - can use higher launch power and higher melting point (so even less amplifiers)
 - Can inject more channels in one fibre
 - Higher GSNR
- It has lower latency: close to speed of light in the vacuum
- Can have very large bandwidth (low loss across wider spectrum)
- Lower dispersion (no material dispersion, so it can be shaped very well with cross section design)

Disadvantages

- High fabrication complexity (due to precise complex geometry)
- Building adequate splicing and connectors
- Possible durability issues (thin complex glass structure)

Mostly related to fabrication, but still very early stage



Petrovich, M., Numkam Fokoua, E., Chen, Y. et al. Broadband optical fibre with an attenuation lower than 0.1 decibel per kilometre. Nat. Photon. (2025). <https://doi.org/10.1038/s41566-025-01747-5>